

Architectural Justification: Student Project Portfolio & Showcase App

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Architecture Selection: "We chose Client-Server Architecture for the Student Project Portfolio & Showcase App."

- Architectural Choice:** We selected a 3-Tier Client-Server Architecture consisting of a Presentation Tier (web portals for students, judges, and public visitors), an Application Services Tier (API Gateway, Submission Manager, Rubric Evaluation Service, and Voting & Anti-Sybil Engine), and a Data & Storage Tier (Real-time Leaderboard Aggregator, Media Repository, and PostgreSQL datastore).
- Reason 1 (Centralized Authority & Scoring Integrity):** A client-server model ensures that rubric grading records from judges and public voting tallies are verified and aggregated on a central server, preventing any client-side tampering with project scores or awards.
- Reason 2 (Asymmetric User Roles & High Concurrency):** The platform supports three distinct concurrent workflows—student team project submissions, panel judge rubric evaluations, and high-traffic public voting—which decouple cleanly through centralized REST and WebSocket endpoints.
- Security Advantage:** The Voting & Anti-Sybil Engine operates behind an authenticated server gateway, applying device fingerprinting, IP rate-limiting, and CAPTCHA validation to reject automated vote stuffing before any ballot touches the database.
- Performance Benefit:** Decoupling the read-heavy Leaderboard Aggregator with an in-memory Redis cache allows live ranking recalculations and WebSocket broadcasts to respond in under 300 ms, satisfying the NFR-001 latency target under peak expo loads.

Component & Interface Specifications (UML 2.5 Mapping)

Component Name	Tier / Layer	Stereotype	Provided Interface	Required Interface	Key Responsibility
API Gateway & Auth	Presentation	«component»	IExpoClientAPI	ISubmissionService, IEvaluationService, IVotingService	Routes requests, terminates TLS 1.3, verifies JWTs, and enforces rate limiting.
Submission Manager	Application	«component»	ISubmissionService	IMediaStorage, IDataStore	Manages project profiles, abstracts, GitHub repository URL verification, and demo video links.
Rubric Evaluation	Application	«component»	IEvaluationService	ISubmissionService, IDataStore	Renders digital rubric forms, validates judge criteria scoring, and locks completed evaluations.
Voting Engine	Application	«component»	IVotingService	ILeaderboardService, IDataStore	Validates public voter authenticity, checks device fingerprints to prevent Sybil attacks, and queues ballots.
Leaderboard Service	Data / Cache	«component»	ILeaderboardService	IDataStore	Merges judge grades and public votes into live rankings with sub-300 ms Redis response latency.
Media Repository	Data / Storage	«component»	IMediaStorage	Cloud Storage API	Coordinates multipart uploads of posters and slides with presigned URLs and virus checks.